DAY 07 | PRE-READ

Connections, Promises and Manuals

A light read before Day 7. About 15 minutes, no code required.

Prepared by **Ahmed** | linkedin.com/in/ahmed-al-imaad

Why read this?

Real data is connected. Customers place orders, orders contain products, products belong to categories. Tomorrow you will connect tables, make sure changes are safe, and write documentation that others actually enjoy reading.

1. How things relate

Relationship	Everyday example
One-to-one	One person has one passport.
One-to-many	One teacher teaches many students in a class; each class has one teacher.
Many-to-many	Students join many clubs, and each club has many students.

For many-to-many, you need a register in the middle: a sign-up sheet that lists which student joined which club. Databases use exactly this idea, called a link table.

2. All or nothing

When you transfer money, two things must happen: your account goes down and your friend's goes up. If only the first step happened, money would vanish. Databases protect you with **transactions**: a group of steps that either all succeed or are all undone.

> **Key idea:** commit means "make it permanent". Rollback means "undo everything since we started".

3. A manual that is always up to date

Have you ever followed a user manual that described an older version of a product? Frustrating. FastAPI builds its documentation directly from your code, so it is always current. The standard behind it is called **OpenAPI**, and the interactive page you have already seen at /docs is called **Swagger UI**.

Good documentation answers three questions quickly: what can I do, what must I send, and what will I get back?

4. Think about these before class

- List three relationships in an online shop and classify each as one-to-one, one-to-many or many-to-many.
- Describe a situation in your job where a half-finished change would cause problems.
- What makes a manual or guide easy to follow?

Tsunagari (繋がり): connection, the links between things